



ASSESSMENT 1 BRIEF

MODULE CODE	COM4113
MODULE TITLE	Tech Stack
MODULE LEADER	Dr Yashar Baradaranshokouhi
ASSESSMENT TITLE	Tech Stack Proposal
WEIGHTING	30%

ASSESSMENT LEARNING OUTCOMES

Upon successful completion of this assessment, you will be able to:

1. Apply relevant tools and techniques to design and develop a web application, justifying the choices made.
2. Analyse constraints and opportunities afforded by different programming techniques, frameworks or libraries, and choice of tools in the production of a modern web application.
3. Apply modern industry standards and take into consideration the security, ethical, legality, usefulness, and usability of the finished applications as well as development efficiency.

INSTRUCTIONS

For this assessment, you will develop an online platform called the **"Community Skill Swap Hub"**. This platform will be a space for users to share their passions, hobbies, and skills with a community of like-minded learners.

The core idea is that a user passionate about a hobby (e.g., Urban Gardening, 3D Printing, Baking, or Coding) can create a profile and post links to their favourite online tools, books and other resources. This includes AI tools, books, blogs, articles, products/services, videos, and websites. These resources will be organised into different categories, which will be the hobbies themselves.

You are tasked with choosing an appropriate tech stack, including a server-side language and framework, and user interface frameworks/toolkits. You will then create a basic implementation that showcases the main features to run as a pilot test for the idea. From this initial version, the application must be designed to be easy to grow and scale up.

A MINIMUM VIABLE PRODUCT is provided with a list of requirements for the project. This should not be considered the minimum work that you should complete to pass the module. You should read the criteria in the MARKING CRITERIA and meet the criteria for a 3rd mark (40%) to pass the assessment. You will only progress to higher marking category if you have met the criteria in the previous category. Please refer to the marking criteria for a comprehensive list of criteria and specifications for each criterion.

WHAT YOU NEED TO DO

Here are some steps to guide you through your project.

1. Analyse the problem you have been given, identify and document a range of previous works most relevant to the current project.
2. You are developing the front-end only, however, you should read about different technology stacks for developing the back-end as part of the rest of the project (to be continued in the second semester), and write about your choice of the back-end technologies.
3. Select the front-end technologies and frameworks that you think will be appropriate for the user interface.
4. Implement the user interface. Since there is no database required for this part (UI), the interface should contain mock data.



5. Develop a project plan where you describe the technologies you have chosen and how you will use them to complete the project. The project plan should include a comprehensive list of tasks and a schedule of delivery.
6. You should access and accept the GitHub classroom repository for the assessment, and you should submit a link to your active Git repository containing your project code.
7. You should use the following GitHub Classroom repo for the purpose of the assessment <https://classroom.github.com/a/CRqtaUTR>.
8. You **MUST** submit an archived version (.zip file) of the same repository to the assessment portal on Moodle. The zip file should contain all components of your work, as well as the documentation. Only the submission to Moodle will be marked.
9. Your project repository should contain a ReadMe.md file with basic installation and usage instructions, requirements, and the required information to be able to reproduce your development environment and run your application.

MINIMUM VIABLE PRODUCT (MVP)

The MVP should be fully functional in terms of the core features described. It should be free of critical bugs that prevent users from creating profiles, searching, or browsing. User data should be handled securely, especially authentication and personal information.

USER REGISTRATION and AUTHENTICATION

- Users can sign up, log in, and manage their accounts.
- Simple login page with email address and password.

USER PROFILE

- User profiles should include the following fields:
 - Full Name (mandatory)
 - Contact Information
 - Email (mandatory)
 - Pronouns (optional)
 - Phone number (optional)
- **Photograph Upload**
 - Users can upload a profile picture (image file upload mandatory).
 - Users can update their profile photo (delete and replace their photo).
- **Their Hobbies or Skills**
 - Users should provide their hobbies or skills (e.g., art, music, web development, baking, sports, etc.).

COMMON FEATURES

- Drop down menu to help users easily navigate and find specific resources/tools.
- An "About Us" page where the reader can learn more about the purpose of the **Skill Swap Hub**.
- A "Contact Us" page where users can contact the customer support team.
- Resources and tools should be categorised by hobby (e.g., Coding, Baking, Art, AI in Hobbies, etc.)
- Add a page where users can post resources (links to tutorials, articles, or GitHub repositories).
- Each post should include a title, description, category, and link.
- Basic search functionality to find resources or tools by keywords.
- Filter by category (hobby) to narrow down relevant resources/tools.
- **User Feedback**
 - Basic upvote or rating system for resources and tools to indicate their usefulness.
 - A simple review or comment system where users can leave feedback on shared resources (e.g., "This tutorial was fantastic, thanks for sharing!").
- The application should be usable on both desktop and mobile browsers.



- A simple homepage that directs users to either **share a resource**, **create a profile**, or continue as a guest user to explore and browse hobbies, pages and learn more about the website.
- All mandatory fields should be displayed with star sign (*) on the portal.
- Users should be able to search for tools.
- Users should be able to view the tools in categories.
- A section specifically for **AI tools that are relevant to hobbies** (e.g., AI for art generation, AI for music composition, or AI for code optimization).

DOCUMENTATION

Create a ReadMe.md file consisting of the following sections:

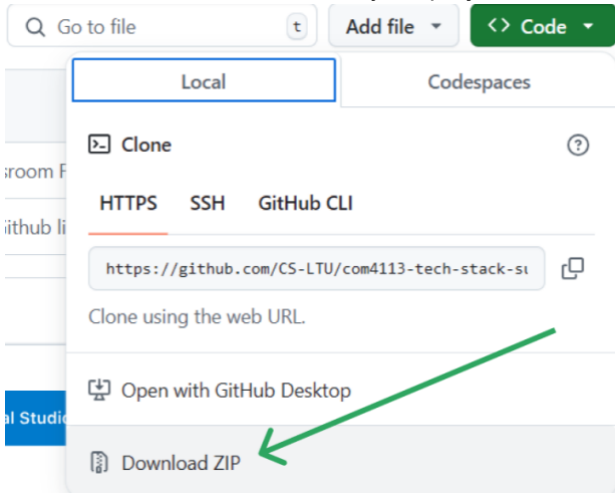
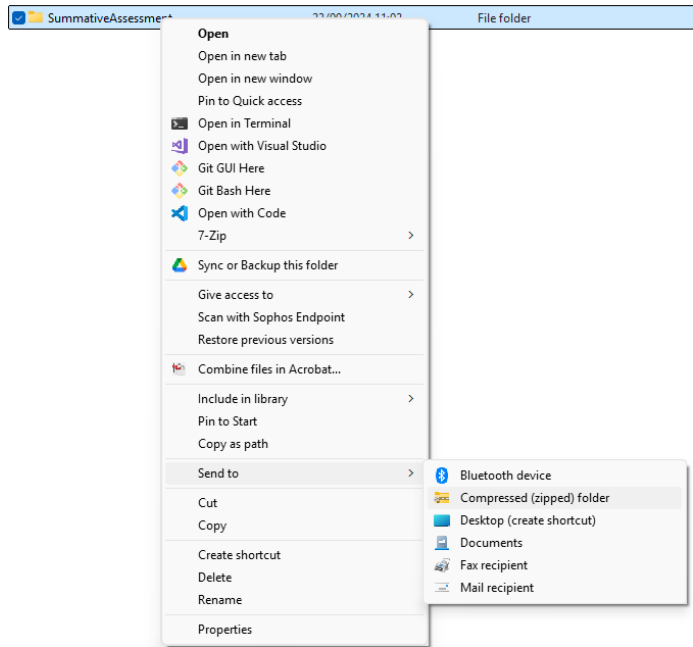
- Choice of tech stack
 - Identify and discuss your choice of tech stack for developing the project.
- Version control
 - Access and accept the invite to [designated GitHub classroom repo for the assessment](#) and include a link to your GitHub repository in your ReadMe.md file with a short description of how you have utilised Git version control.
- Project overview
 - Project goals and objectives.
 - Scope of the project.
- Installation instructions
 - A step-by-step guide that explains how to set up the development environment, how to install dependencies, and how to get the application running on a new deployable environment.
 - List of packages and libraries required.
 - Possible errors and technical challenges that you faced in developing the project.
 - How did you overcome these technical challenges.
- Project plan
 - Timeline of developing features/requirements.
 - Front-end/user interface.
 - Back-end/server-side.
 - Other milestones.
 - Ensure that your project plan is detailed and realistic, accounting for the scope of the project and your academic calendar.
 - Adjustments to the plan may be necessary as the project progresses, and such changes should be documented.
- User journey
 - How is the user is expected to navigate the website.
- Legal and ethical considerations
- Risk assessment
- Real-world applications of the platform
- Good Software Attributes
- Future considerations for scaling
- In-code documentation for key functions and components
 - Add comments in your code to make it easier for other developers to read and work with your code in the future.

OPTIONAL ENHANCEMENTS (POST-MVP)

- User profiles: Let users showcase their shared resources, history, and ratings.
- Resource bookmarking: Allow users to save resources to their personal collection.
- AI integration: Include an additional page to make automated recommendations for AI tools based on the user's activity or interests.
- Advertise and request for tools in a specific category
- Voting system to support a request for specific category of trainings/skills



SUBMISSION DETAILS

RELEASE DATE	20 th November 2025
SUBMISSION DATE	Friday 12 th of December 2025 at 12:00 PM (noon)
DELIVERABLES	You should submit an archive file (.zip) including all of your work. If you have synced the latest version of your work to GitHub, you can download an archive format of your project from GitHub.  As an alternative, make sure that all of your project files and corresponding documentation is stored in a folder within your machine (laptop/VM/PC) and compress the folder as an .zip file.  Upload the .zip file to Moodle. Only submissions on Moodle are eligible for marking.
SUBMISSION DETAILS	Submit your assignment by uploading it to Moodle <u>before midday</u> on the submission date. This deadline will be automatically and strictly enforced. If your submission is late, your grade may be affected. If you have any issues submitting your work, you must email the assessment team and copy in the module leader <u>before the assessment due time</u>. Do not leave



	your submission until the last minute to avoid any penalties due to problems with the submission portal. Assessment Team: assessment@leedstrinity.ac.uk Module Leader: yash.b@leedstrinity.ac.uk We may ask for a demonstration of your work following the submission. If needed, this will be communicated to you individually via email. Please check your emails regularly.
--	--

Your feedback / feed forward and mark for this assessment will be provided within 15 working days.

MARKING CRITERIA

Marks are awarded based on the following criteria. Within each part, aim to complete the work for each section before moving on to the next. The following banded marking scheme is used:

<i>Exceptional 1st</i>	100/95/92		<i>2:ii</i>	58/55/52
<i>Outstanding 1st</i>	88/85/82		<i>3rd</i>	48/45/42
<i>1st</i>	78/75/72		<i>Bare Fail</i>	38/35/32
<i>2:i</i>	68/65/62		<i>Fail</i>	25/20/10/0

If you have completed all the preparatory exercises and attended your classes, the estimated additional time required to PASS this assessment is approximately 30 hours.

To obtain a 3 rd mark (40%), you must have:	○ Provided a markdown file (ReadMe.md) with generic discussions that covers basic ideas. ○ Delivered most of the Error! Reference source not found. with a justification for any missing features. ○ Demonstrated a foundation level of technical skills and knowledge in the documentation and artefact. ○ A product that is functional with minimal debugging. ○ A basic project timeline and list of key milestones. ○ Included minimal comments on the choice of frameworks and tech stack for the development of the web application. ○ Discussed legal and ethical considerations in a basic manner and carried out a generic level of risk assessment of the project. ○ Submitted evidence of how you have utilised version control and make project files available on GitHub.
To obtain a 2:ii mark (50%), you must have (in addition to the above):	○ Documented a rationale and justification for the choice of tech stack and technologies on the server-side, front-end and back-end development. ○ Provided at least one example of real-world applications. ○ Referencing with minor inaccuracies or inconsistencies. ○ Submitted a ReadMe file that holds some instructions and guidelines on how to replicate the steps to create and run the project. ○ Developed and submitted a project that complies with no errors and meets the requirements specified in the MVP. ○ Demonstrated a consistent naming convention applied across the code in functions and variables. ○ Included a visual presentation of the project timeline (Gantt Chart) with evidence of tracking progress on the milestones.
To obtain a 2:i mark (60%), you must have (in addition	○ Demonstrated a good level of competency and technical skills in implementing and utilising intermediate level of web development tools and techniques.



to the above):	○ Compared and critically discussed the rationale and justification for choice of tech stack framework, server-side, front-end and back-end development. ○ Applied and documented your knowledge and understanding of good software attributes in the design and development of the project artefact. ○ Commented the code, leading to better understanding of developer's logical thinking and any challenges resolved in the code. ○ Critically discussed the legal and ethical considerations. ○ Delivered a high-quality semi-professional level product exceeding the MVP requirements and delivered at least one of the optional enhancements (POST-MVP).
To obtain a 1st mark (70%), you must have (in addition to the above):	○ Demonstrating a tabular comparison of the previous works, identifying gaps, and features in addition to those in the MVP. ○ Provided a critical discussion around the current gaps in the applications/platform and how the future works can contribute to this. ○ Developed and documented at least three of the optional enhancements (POST-MVP on page 3). ○ Included a ReadMe file that holds comprehensive instructions and guidelines on how to replicate the project structure, install required dependencies, and packages and execute the project. ○ Submitted your reflections including changes to the project timeline and challenges you have faced in completing the project. ○ Included a visual presentation of the site map for the website in the documentation (designed in Figma or use of flowchart) ○ Referenced at least 10 quality resources in the correct format. ○ Submitted a comprehensive risk assessment, detailing the identification of the risk factors and effective strategies to contain them within the scope of the project. At least two of the risk mitigation strategies should be implemented and documented.

USE OF GENERATIVE ARTIFICIAL INTELLIGENCE IN THIS MODULE

You may use generative AI such as Microsoft Copilot to assist you in the process of undertaking the assessment in the following ways: brainstorming, research, planning, feedback, editing.

All use of generative AI must be explicitly acknowledged, and any artificially generated content (e.g. images) explicitly labelled, with the source of the AI tool referenced using current APA referencing conventions. You can find further guidance on the library website on their AI webpage).

In submitting your assignment, you agree to disclose the extent to which you have used generative AI in preparing this work and include evidence of your AI use in your appendices (e.g. dated screen shots of your use of this tool or copy and paste your AI chat into Word).

Failure to disclose your generative AI use may result in a zero for your assignment and a referral for academic misconduct (see the Student Academic Misconduct Policy under Essential Info in the MyLTU app).

Include one of the following statements on your assignments.

Either:

This assignment used generative AI in the following ways for the purposes of completing the assignment (choose 1 to 5 of the following): brainstorming, research, planning, feedback, editing.

Or:

This assignment did not use generative AI for the purposes of completing the assignment.



ACADEMIC MISCONDUCT

Academic Misconduct includes all forms of academic dishonesty, whether intentional or accidental, that compromise the integrity of the University's assessment processes. It is essential that you review our [Student Academic Misconduct Policy](#) to understand the guidelines and the serious consequences that may arise if they are not followed.

HELP AND SUPPORT

- Please use the module handbook and the [Computer Science Community Teams site](#) as a source of information. Do try and find the answer out yourself before reaching out for help.
- Support will be provided via Microsoft Teams and email during office working hours. You can also ask questions during your timetabled sessions. You may request a one-to-one meeting with a tutor during their published office hours.
- The Student Support team are available seven days a week to support you in all aspects of student life. This could be for support relating to your course, your accommodation or for more general advice such as relationships or your wellbeing. Log in to the LTU app to access support services.
- The full set of university guidelines on assessments, deadlines, and extensions is available on the LTU app, please familiarise yourself with the documentation.